MCS-CEV Optimization Framework Description and Summary (Draft)

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I. INTRODUCTION

The purpose of this Agreement is to fund the demonstration and optimization of Mobile Charging Stations (MCSs) for charging construction electric vehicles (CEVs). This research will demonstrate the potential to reduce air pollutants by upgrading heavy-duty construction equipment from internal combustion engines (ICEs) to CEVs while fulfilling the construction tasks. It will quantify reductions in carbon emissions, operating costs associated with fuel and electricity use, and air pollutants. In addition, the study will evaluate grid-level benefits from using MCSs, including reduced peak demand and the ability to defer or avoid distribution grid upgrades, while still meeting all charging requirements.

The goals of this Agreement are to:

- Determine the optimal location, timing, and rates for CEVs charging and MCSs re-charging, subject to the construction schedule, power/energy capacity of CEVs and MCSs, and other physical constraints, including the substation peak power capacity.
- Prove the benefits of CEVs over conventional internal combustion engine (ICE) construction equipment in terms of reduced operation costs, lower carbon emissions, and decreased air/noise pollutant levels.
- Prove the benefits of MCS over current fixed charging stations (FCSs), in terms of peak demand reduction, deferred grid upgrade costs, and reduced charging time.
- Prove the benefits of MCSs for grid services including but not limited to increased utilization of existing EV charging stations, higher renewable self-consumption, and emergency building services by integration with the world-renowned UCSD microgrid.

The objectives of this Agreement are to:

- Validate operations of MCSs for CEVs charging under regular construction activities for at least six months at multiple construction sites;
- **Reduce Costs and Emissions:**
    - Demonstrate at least 10% peak demand reduction and charging time reduction (including travel, waiting & actual charging time) by using mobile charging stations versus the fixed charging station benchmark for construction equipment charging.
    - Demonstrate at least 50% carbon emission reduction by using electric construction equipment versus internal combustion construction equipment.
- **MCS-to-Grid Benefits:** at least 5% increase in charging station utilization and renewable self-consumption, and validation of MCSs for building emergency services.
- **Technology Transfer:** Test the MCS-to-CEV solution with different use cases, including but not limited to different MCSs battery types, CEV types, and construction locations.

This report captures the framework, methodologies, results, and conclusions of the general optimization framework for charging the CEVs and recharging the MCSs. **The optimization framework decides *when*, *where,* and by *how much* the MCS should serve the CEVs.** The optimization framework

minimizes MCS charging costs, peak power demand from the grid, carbon emissions, and construction job delays while enforcing the MCS and CEV battery power and energy limits.

The project provided an opportunity to develop the spatio-temporal MCS-CEV optimization framework, layout the different operation strategies for the MCS-CEV operation at construction sites, and helped us simulate and analyze the potential cost, emissions, peak demand, and charging time reduction benefits of the MCS-CEV framework, and perform parametric analyses across two realistic UC San Diego construction sites.

The main outcomes of this project are to **elucidate the feasibility of using MCS as a replacement for fixed charging stations (FCS) for CEVs recharging**, and more broadly to **use MCS-CEV instead of conventional diesel-powered internal combustion engine (ICE) construction equipment.**

II. OPERATING STRATEGIES OF THE CONSTRUCTION EQUIPMENT

We operate the construction equipment based on four operating strategies. Table 1 gives a high-level summary of the various operation strategies.

A. Using ICE-powered construction vehicles - No MCS/FCS/CEVs

The first operation strategy does not involve the use of FCS, CEV, and MCS. This strategy involves the use of conventional ICE construction vehicles, which operate on traditional diesel fuel.

B. Charging CEVs directly from the grid through FCS - No MCSs

The second operation strategy does not involve the use of MCSs; instead, each construction node has an FCS for recharging the CEVs. Each of the FCS has two ports for charging the CEVs. **The CEVs, after a full overnight charge, commence with the day's work till running out of charge (possibly during scheduled construction work). Once it runs out of charge, it goes to its FCS node to recharge its battery with full power till it can again work**, as there is a huge penalty for missed work (as any missed work has to be made up by extra equipment and crew hours later). After recharging, the CEV comes back to work until it again runs out of charge and repeats the process, or till the end of the day-shift.

The CEVs are not allowed to travel from one construction node to another, as CEVs are highly immobile and moving them for charging between construction nodes during construction risks injuries and causes hindrance to commuters. After the time of work for the day is over, the CEVs recharge to full battery from the FCS immediately. This is because of two reasons: (i) To prepare for the next day's operations, and (ii) To minimize the CEV operating crew's hours of work. Note that if both the plugs of FCS are occupied, each CEV is charged to half of the full power rating of the FCS.

C. Charging CEVs from the MCSs, which recharge from the grid without optimal scheduling

The third operating strategy does not involve the use of FCS; instead, the CEVs are recharged through the use of MCS, which are in turn recharged from the grid.

Each of the MCSs has two ports for CEVs recharging. **The CEVs are recharged till it can again work if it runs out of charge during operations, as in operating strategy B**. Here, the MCS can travel from one node to another, with the first priority for CEVs at the current node of the MCS. **After the day's work is over, the CEVs recharge from the MCS to full charge. Finally, the MCSs are recharged from the grid at the grid connection nodes to full charge at full power to be ready for the next day's work.**

D. Charging the CEVs from the MCS, which charges from the grid with optimal scheduling

Here, the operation strategy is similar to that of Section II.C, with the major difference being that **the time, location, and rate of recharging of MCS from the grid and recharging of the CEVs from the MCS is modulated by our proposed optimization algorithm, which minimizes the penalty for lost work of the CEVs, electricity costs (both energy and demand charges) and emissions due to MCS recharging from the grid.** The optimization algorithm thus decides when and where and by how much to charge the CEVs from the MCS, and also the MCS from the grid.

Table 1: Comparison of different operating strategies of the construction equipment

Operation Strategy	FCS present	MCS present	MCS optimally scheduling
A. Using ICE-powered construction vehicles	No	No	NA
B. Charging the CEVs directly from the grid through FCS - No MCS	Yes	No	NA
C. Charging the CEVs from the MCSs without optimal scheduling	No	Yes	No
D. Charging the CEVs from the MCS with the proposed optimization framework (Ours)	No	Yes	Yes

Operation strategy A does not involve any optimization or involvement of FCS/CEV/MCS, as it is based on conventional ICE-operated construction equipment. Operation strategies B and C are other non-optimization-based baseline cases, with limited flexibility in charging of CEVs and/or MCSs.

III. OPTIMIZATION FRAMEWORK

To meet the CEV charging demands in a specific area, a CEV charging framework consisting of MCSs, depots for MCS charging (grid connection nodes), and construction sites (i.e., construction nodes) is proposed in this report, as depicted in Fig. 1. The system operator is assumed to be the owner of the MCSs in the considered area, which are towed by electric pickups from one node to another, and the CEVs in this area are charged through the MCSs. Note that for simplicity, we assume that these eclectic pickups are themselves powered by the MCS.

The MCS-to-CEV charging optimization determines the optimal locations, timings, and rates for charging CEVs and recharging MCSs to minimize carbon emissions, reduce MCS charging costs, reduce peak loads, and reduce loss of construction work, taking into account the construction schedule and the power/energy capacities of MCSs and CEVs, as shown in Fig. 2.

The complete mathematical framework for the spatio-temporal optimization (Operation Strategy D in Section II.C) is provided in the Appendix. To maintain brevity and address space constraints, we do not present the full optimization formulation in the main text. Instead, the main text focuses on conveying the key ideas and design considerations behind the optimization approach.

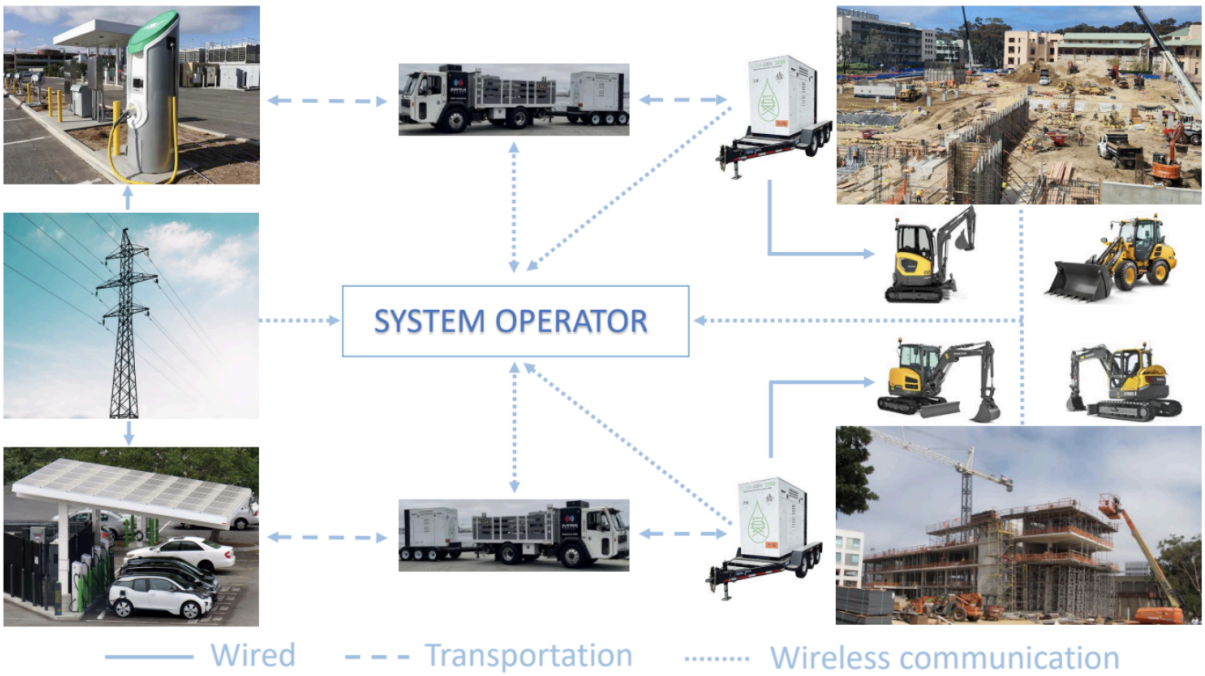

Fig. 1. Conceptual overview of the proposed MCS-to-CEV system.

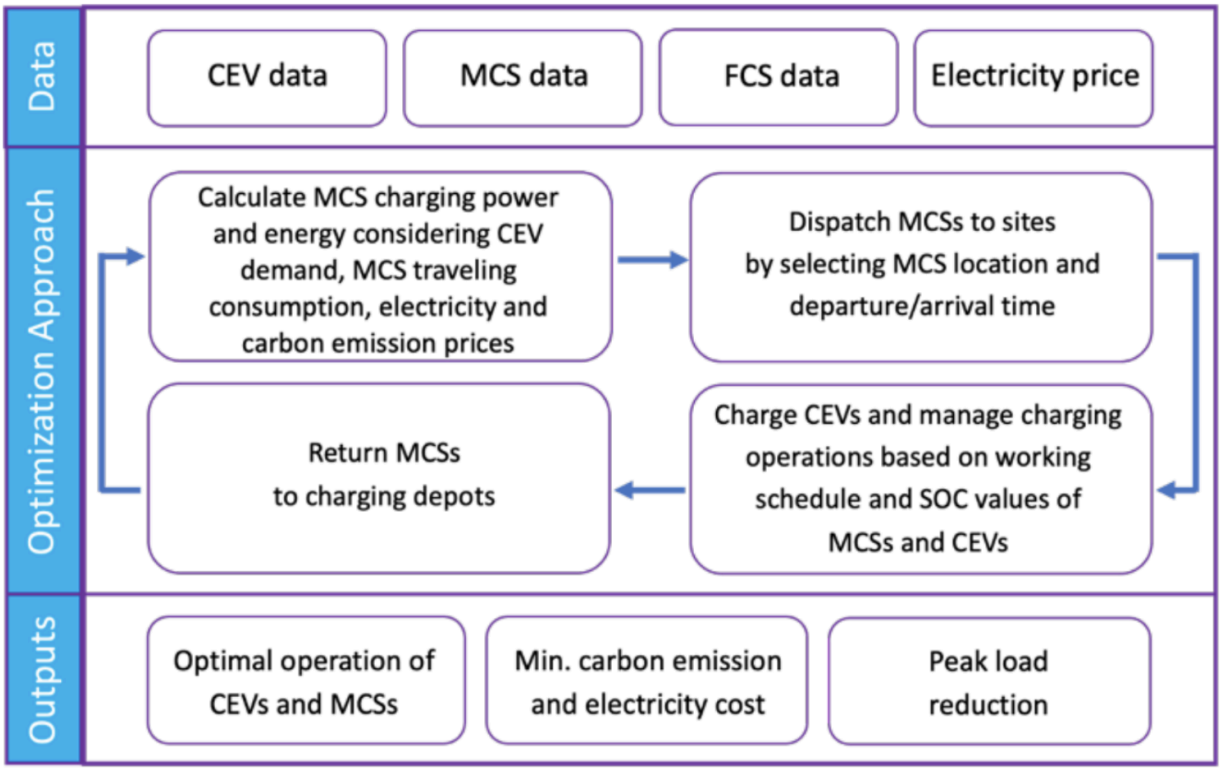

Fig. 2. Schematic representation of the proposed optimization algorithm.

A. Objective Function

The objective of the proposed framework is to minimize the total carbon emissions, electricity costs, and peak demand charges for the MCS recharging from the grid as outlined in the first, second, and third terms in (1), respectively. The objective function also minimizes the missed construction work due to recharging of the CEV during scheduled construction hours, as represented by the fourth term in (1). The objective function is formulated as,

$$\underset{\{P^{\mathrm{ch,tot}}_{m,i,t},P^{\mathrm{dch,tot}}_{m,i,t}\}_{(m,i,t)\in M\times N\times T}}{\text{minimize}} \sum_{m,t} P^{\mathrm{ch,tot}}_{m,t}(\lambda^{\mathrm{CO_2}}_t + \lambda^{\mathrm{buy,elec}}_t)\Delta T + \lambda^{\mathrm{dc}} \max\Big\{\sum_m P^{\mathrm{ch,tot}}_{m,t}\Big\}_{t\in T} + \rho^{\mathrm{miss}} \sum_{i,e,t} P^{\mathrm{missed}}_{i,e,t} \tag{1}$$

where, $\lambda^{\mathrm{CO_2}}_t$ is the penalty due to carbon emissions ($/kWh), $\lambda^{\mathrm{buy,elec}}_t$ is the electricity costs from the grid ($/kWh), λ^{dc} is the demand charge rate from the grid ($/kW), while $P^{\mathrm{missed,work}}_{i,e,t}$ represents the missed work (kW) for each CEV at construction node i due to the recharging activities, with ρ^{miss} being a large weight or penalty constant ($/kWh) introduced to allow missed works only when absolutely necessary.

Essentially, the framework seeks to identify the optimal intersection of minimal carbon emissions and electricity costs and peak demand for the MCS charging, while adhering to the construction working schedules as closely as possible. Refer to the Appendix for the notation.

B. Constraints

See the Appendix for the detailed formulation of the various constraints in the optimization framework. Broadly, for the **charging and discharging power constraints of the MCSs and CEVs**, we incorporate constraints such as: (i) charging and discharging power limits of the MCS, (ii) charging and working power limits of the CEV, (iii) limiting the charging and discharging locations of the MCSs and CEVs which allow charging of the CEVs (or equivalently discharging of the MCS) only at the construction nodes, and charging of the MCSs only at the grid connection nodes, (iv) limiting the duration of work of the CEVs only to the scheduled construction hours, and finally, (v) formulation of the missed work during scheduled construction hours.

For the **energy consumption and state-of-energy (SOE) constraints of the MCSs and CEVs**, we incorporate constraints such as: (i) formulating the SOE dynamics of the CEV and MCS, (ii) formulating the energy requirement of the MCS in traveling from one node to another, (iii) limiting the SOE the CEV and MCS between predefined limits to preserve the battery health, e.g., between 20%-100%, and, (iv) limiting the final and initial SOE of both the MCS and CEV to be the same as their initial SOE to ensure that there is no net charging/discharging over the day.

For the **spatial movement and connection status of the MCSs**, we incorporate constraints such as: (i) limiting the number of connections to an MCS for CEV charging to the number of available plugs on the MCS, (ii) avoiding the visitation of multiple nodes at the same time by the MCS, (iii) avoiding the arrival and departure of the MCS at the same node at the same time, and, (iv) limiting the number of visits by the MCS to a node to once per day to mitigate congestion issues and avoid unnecessary traffic.

IV. CASE STUDIES

Here we carry out case studies based on the four operating strategies introduced in Section II. Operation strategies A, B, and C are baseline strategies which we compare our optimal scheduling (Operation strategy D) against.

Operation strategy A using conventional ICE construction equipment (such as a diesel powered JCB mini excavator and JCB telehandler) assumes the use of diesel fuel with a specific fuel consumption, $SFC = 0.22$ kg diesel/kWh, specific gravity $\rho_{\text{diesel}} = 0.85$kg/l, and conversion factor of liters of diesel to kgCO_2 as $CF = 2.6569$ kgCO_2/l. Thus, the emissions of CO_2 per kWh of construction work is given by,

$$\frac{SFC \cdot CF}{\rho_{\text{diesel}}} = 0.688 \quad \text{kgCO}_2\text{/kWh.}$$

The cost of diesel is assumed to be $4.9/gallon. Using the conversion factor that 1 gallon corresponds to 0.264 liters, the operating cost of ICE construction equipment per kWh of construction work is given by,

$$\frac{SFC \cdot 0.26}{\rho_{\text{diesel}}} \cdot 4.9 = 0.335 \quad \text{\$/kWh.}$$

Each of the operation strategies is investigated under two broad simulation scenarios (see Section IV. A) reflecting realistic UC San Diego construction sites and layouts, which take into account different **(i) number, and (ii) type of CEVs and MCSs; and (iii) working schedules of CEVs.**

A. Simulation scenarios and input data

Simulation Scenario 1 considers a single construction site where construction activities are performed by two medium-duty construction electric vehicles (CEVs): **a JCB mini excavator and a JCB telescopic handler.** For Operation Strategy B, the site is equipped with a single Level-2 fixed charging station (L2 FCS), similar to a ChargePoint CP6000. For Operation Strategies C and D, the fixed charging station is replaced by an MCS, specifically a Saniset J250.

Simulation Scenario 2 considers two construction sites. The first site has the same configuration as in Scenario 1. The second site includes two heavy-duty CEVs—a heavy-duty excavator and a telescopic handler—and is additionally equipped with a DC fast charging fixed charging station (DCFC FCS), similar to a ChargePoint Express 250. The work schedules of the CEVs at the second site follow the same structure as those at the first site, but with different work demand levels. This scenario includes two Saniset J250 MCSs, which can be dispatched and moved between the two construction sites.

Simulation Setup and Assumptions

- The simulations are conducted over a single day, starting at 00:00 h and ending at 23:59 h. All CEVs and MCSs are initialized at 100% SOE level.
- To preserve battery health, the minimum SOE for both CEVs and MCSs is constrained to remain above 20% throughout the simulation. Each MCS is equipped with two charging plugs, allowing multiple CEVs to be charged simultaneously within the same time interval.
- Each MCS can be recharged at a grid connection node via a preinstalled DC fast charging fixed charging station (DCFC FCS) located 0.1 miles from each construction site. The two construction sites are separated by a distance of 0.5 miles.
- At both sites, excavators operate from 8:00 h to 11:00 h and from 12:00 h to 17:00 h, with a lunch break from 11:00 h to 12:00 h that provides an opportunity for charging. The telescopic handlers operate from 12:00 h to 17:00 h at each site.
- The technical specifications of the MCSs and CEVs are summarized in Table I and Table II, respectively. The network topology is illustrated in Fig. 3. The working hours and time-varying power demands required to complete construction tasks are shown in Fig. 4.

- The local marginal electricity price and carbon emission data obtained from the California Independent System Operator (CAISO) for the considered day are given in Fig. 5. The demand charge rate and the penalty for missed work are considered as $\lambda^{\mathrm{dc}} = 25$\$/kW and $\rho^{\mathrm{miss}} = 100$ \$/kWh. Carbon emission prices for the entire day were derived using the social cost of carbon, estimated by the U.S. government at \$50 per metric ton of CO_2.

Table 2: Technical specifications of the MCS

Name of the MCS	Battery energy capacity, $\mathrm{SOE}_{\mathrm{max}}^{\mathrm{MCS}}$ (kWh)	Charging and discharging efficiencies, η^{ch} and η^{dch} (%)	Maximum charging/discharging power, $\mathrm{CH}^{\mathrm{MCS}}$/ $\mathrm{DCH}^{\mathrm{MCS}}$ /$\mathrm{DCH}^{\mathrm{MCS,plug}}$ (kW)	Road energy consumption , $D_{i,j}$ (kWh/mile)	Noise Level (dB)
Saniset J250	250	95	105/80/40	0.5	55

Table 3: Technical specifications of the CEVs

CEV index and type	CEV 1: Medium duty mini excavator	CEV 2: Medium duty telescopic handler	CEV 3: Heavy duty excavator	CEV 4: Heavy duty telescopic handler
Quantity	1	1	1	1
Maximum charging power (kW)	8.8	8	18	15
Battery energy capacity, (kWh)	19.8	25	40	45
Working capacity, M (kW)	7	6.8	15	13.7
Site Location	1	1	2	2
Total work requirement (kWh/day)	24	20	55	35
Noise level (dB)	87	93.6	Not available	Not available

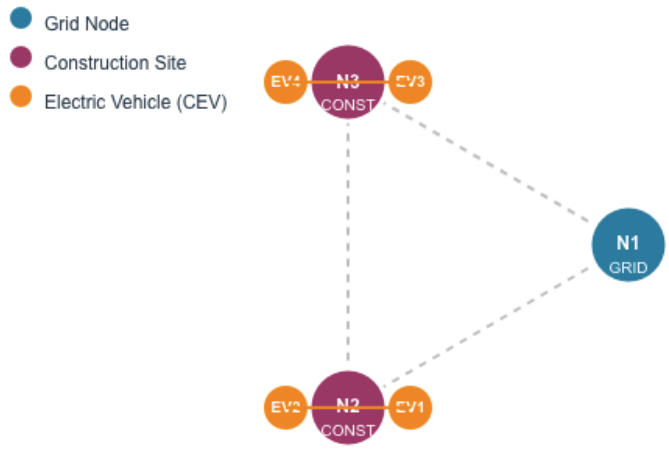

Fig. 3. Network topology with grid nodes (blue), construction sites (purple), and assigned electric vehicles (orange). Dashed lines represent possible connections between nodes.

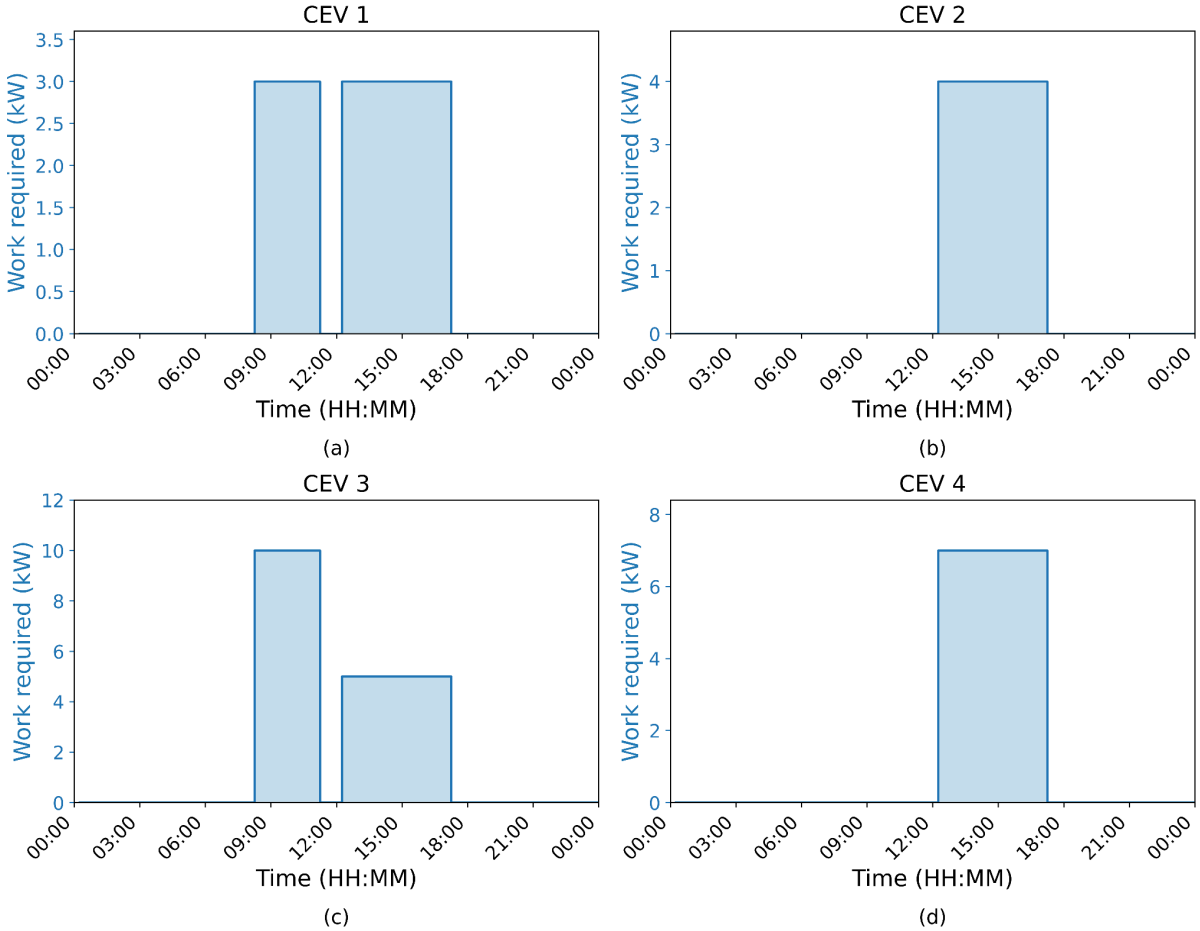

Fig. 4. Required power time series for the 4 CEVs at the 2 construction sites. Note that scenario 1 only has CEVs 1 and 2, while scenario 2 incorporates CEVs 1 through 4

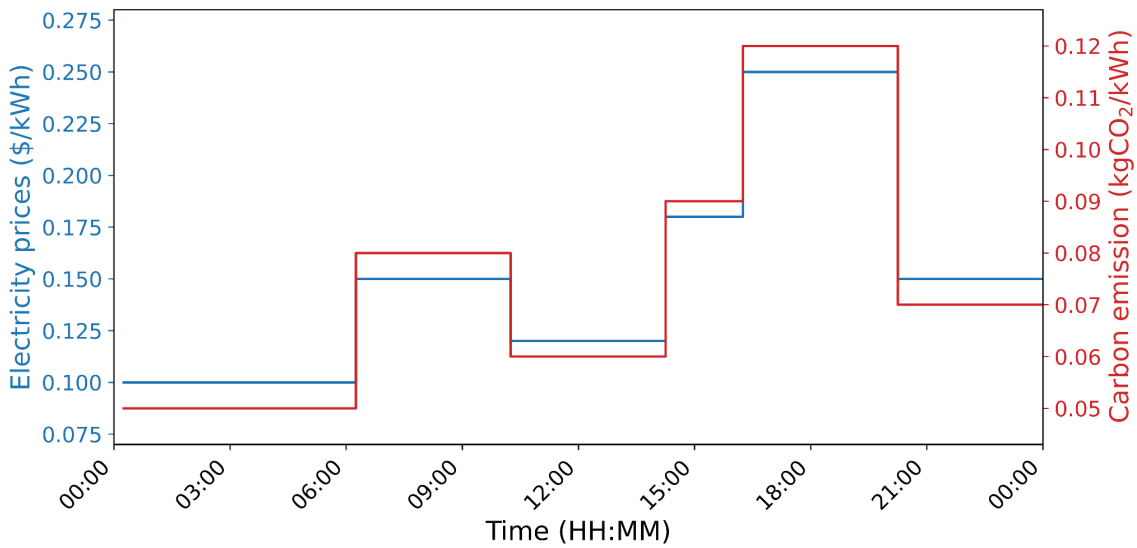

Fig. 5. Daily electricity price and carbon emission time series

V. RESULTS AND DISCUSSIONS

A. Simulation scenario 1

Operation Strategy A uses ICE construction equipment powered by diesel. As a result, it requires no new capital investment, and operating costs consist solely of diesel fuel expenses.

Operation Strategy B introduces additional complexity, costs, and risks for construction projects that start with electric construction equipment, because you must design and install higher-capacity temporary or early permanent power, coordinate with the utility sooner, and potentially arrange off-site or generator-based charging. This introduces more early-phase risks: utility delays, equipment lead times, and charger downtime can idle crews that diesel machines would keep productive. Costs rise due to larger temporary electrical infrastructure, added trenching and concrete, possible utility upgrade fees, and logistics or fuel costs for interim charging solutions. Well-planned projects can mitigate rework by aligning early charging infrastructure with the final EV charger layout so that much of what's installed for construction becomes part of the permanent system. At a minimum, having an on-site FCS requires a new capital investment of approximately $9,500 to purchase and install a dual-port Level-2 charger, such as the ChargePoint CP6000, which provides up to 19.2 kW per port. Additional costs, on the order of $20k to $100k, may be incurred to upsize the temporary power to the site due to added trenching, conduit (1 phase to 3 phase), pads, and gear.

Operation Strategies C and D require no additional capital investment, as the MCSs are charged at pre-existing grid connection nodes equipped with DC fast charging stations.

The operating costs, missed work, carbon emissions, and peak power demand of the 4 operating strategies for scenario 1 are summarized in Table 4.

Table 4: Operational results for simulation scenario 1

Operation strategy	CO_2 emissions (kg)	Energy cost ($)	Peak demand (kW)	Missed work (kWh)
A (ICE)	30.26	14.74	NA	NA
B (FCS without MCS)	4.75	9.86	16.80	0
C (MCS but not optimal)	4.73	9.95	105	0
D (MCS and Optimal)	3.55	7.47	8.59	0

B. Simulation scenario 2

In addition to the first site depicted in scenario 1, scenario 2 involves the use of an additional site with 2 extra larger CEVs and an additional MCS.

As in scenario 1, operation strategies A, C, and D do not involve any new capital investments. Over and above the capital investments in scenario 1, in scenario 2, Operation strategy B involves additional new capital investment on the order of $20,000 to $100,000+ to buy and install a dual-port DCFC charger like ChargePoint Express 250, which has a maximum power output up to 62.5 kW per port, as well as to secure higher temporary power to the site.

The operating costs, missed work, carbon emissions, and peak power demand of the 4 operating strategies for scenario 2 are summarized in Table 5.

Table 5: Operational results for simulation scenario 2

Operation strategy	CO_2 emissions (kg)	Energy cost ($)	Peak demand (kW)	Missed work (kWh)
A (ICE)	91.29	44.48	NA	NA
B (FCS without MCS)	14.32	29.7	49.8	1.25
C (MCS but not optimal)	12.93	27.33	210	1.25
D (MCS and Optimal)	12.53	26.22	17.57	1.25

Overall, for both scenarios, we see an 84-88% drop in CO_2 emissions once we shift from ICE-powered construction equipment to CEVs. The CO_2 emissions are slightly lower for the case with MCS (even without optimal scheduling) than for the FCS. **We observe a 33-49% drop in energy costs for the shift from ICE equipment to CEVs.** The peak demand is greater for strategy C than strategy B, as the MCS has a larger power capacity than the CEVs, and charging it without optimization at full power from the grid adversely raises the peak demand. **For the peak demand, we see 49-65% reduction for optimal MCS charging as compared to FCS charging.**

For scenario 1, all the strategies complete their work, while for scenario 2, strategies B, C, and D run out of charge in the 12:00 - 17:00 h session, and have to recharge before going back to work, resulting in

some missed work. For our case studies, the number of CEVs at a site is assumed to be the same as the number of ports on the FCS, i.e., two.

If there are **more than two CEVs at a site that run out of charge together**, we will observe a congestion in charging in the case of FCS, leading to **queue formation and increased waiting times, which in turn results in increased total time required for charging.** This congestion can also happen for the MCS (which also has two ports for charging CEVs), but if **multiple MCSs** are deployed, which can be **dynamically routed between the construction sites in response to demand**, more than two CEVs can be charged simultaneously at a site. Thus, the **risk of congestion is reduced**, which, over and above **reducing the risk of charging time and queue formation, also reduces the risk of lost work,** which is heavily penalized. However, in our case studies, we do not observe a reduction of charging time as we assume that only 2 CEVs work per site, and thus a single FCS having two ports is able to charge both the CEVs simultaneously - i.e, the setup of our case study obviates the reduction of charging time which we would observe in general.

VI. COST AND BENEFIT EVALUATION

A. Cost Evaluation

The demonstration of MCS at UCSD utilized 2 Saniset J250s (one rented for a week and another for two weeks), and a Moxion MP-75/600. The weekly rental cost of the J250s were $4000, while the MP-75/600 were rented for free. Thus, the cost of renting the MCS was $12000.

Additionally, L2D2 chargers and cables were rented for charging the CEVs from the MCS for two weeks, which involved a rental cost of $450+$650 = $1,100.

The delivery and pickup fee for the MCS, cables, and L2D2 chargers was $6000. Overall, with taxes, the total rental cost for two weeks of the three MCS equipment was $20,871.

B. Benefits Evaluation

<u>System operator:</u>

a. **Significant reduction in peak demand** as compared to FCS when the MCS is optimally scheduling its charging from the grid.
b. **Significant reduction of operational energy costs** as compared to conventional diesel equipment.
c. The CEVs can cause some loss of work, especially if it runs out of charge (or reaches minimum permissible SOE) during construction activity. In this case, **MCS has more flexibility than FCS** in recharging CEVs, as the MCSs can move between the construction sites. The mobility of the MCS between the construction sites means additional CEVs can be recharged, which would have to queue up and wait for charging in the case of occupied FCSs, as CEVs cannot move to another unoccupied FCS at a different site. This loss of work is generally negligible for ICE-powered construction equipment, as diesel refueling is generally assumed instantaneous.

<u>Construction crew</u>

a. **No local exposure to carbon, particulate matter, NOx, THC emissions**, with the effects on the environment at large with respect to carbon emissions still being significantly less than conventional diesel equipment.
b. **Significant reduction of noise and air pollution** at the operator level and out of the machine as compared to equivalent conventional diesel equipment.

VII. PROJECT GOALS AND OBJECTIVES

This section discusses how the goals and objectives were met through this project.

Goals:

A. Determine the optimal location, timing, and rates for CEVs charging and MCSs re-charging, subject to the construction schedule, power/energy capacity of CEVs and MCSs, and other physical constraints, including the substation peak power capacity

The optimization framework is described in Section II, I, with the detailed mathematical formulation presented in the Appendix. The MCS-to-CEV charging optimization determines the optimal locations, timings, and rates for charging CEVs and recharging MCSs to minimize carbon emissions, reduce MCS charging costs, reduce peak loads, and reduce loss of construction work, taking into account the construction schedule, charging and discharging power constraints of the MCSs and CEVs, energy consumption and state-of-energy constraints of the MCSs and CEVs, and, spatial movement and connection status of the MCSs.

B. Prove the benefits of CEVs over conventional internal combustion engine (ICE) construction equipment in terms of reduced operation costs, lower carbon emissions, and decreased air/noise pollutant levels

1. The **operational costs** of the CEVs are **33-49% lower** than those of the diesel-powered ICE construction equipment. The savings are a result of the lower cost of electricity as compared to diesel.
2. The **CO_2 emissions** of the CEVs are **84-88% lower** than those of the diesel-powered ICE construction equipment. The lowering is due to lower marginal emissions of carbon in the (renewable heavy) CA grid as compared to carbon emissions due to diesel.
3. The medium-duty CEV excavator has **10 dB and 6 dB less noise** at the operator station and emissions from the machine, respectively, as compared to equivalent diesel equipment. The medium-duty CEV telescopic handler has **1 dB and 9.4 dB less noise** at the operator station and emissions from the machine, respectively, as compared to equivalent diesel equipment.
4. Use of CEVs instead of diesel-powered ICE construction equipment also leads to the **elimination of particulate matter, THC emissions, and NOx emissions** at the point of use.

C. Prove the benefits of MCS over current fixed charging stations (FCSs), in terms of peak demand reduction, deferred grid upgrade costs, and reduced charging time.

1. The **peak demand reduction for the optimal MCS charging compared to FCS is approximately 49% and 65%** in Scenarios 1 and 2; respectively.
2. As compared to FCS involving installation of an L2 charger, grid upgrades of $20,000 to $100,000+ are avoided by the MCS. In case of installation of a DCFC charger, grid upgrades of $80,000 to $200,000+ are avoided by the MCS.
3. If there are **more than two CEVs at a site that run out of charge together**, we will observe a congestion in charging in the case of FCS (which has two ports for charging CEVs), leading to **queue formation and increased waiting times, which in turn result in increased total time required for charging.** This congestion can also happen for the MCS (which also has two ports for charging CEVs), but if **multiple MCSs** are deployed, which can be **dynamically routed between the construction sites in response to demand**, more than two CEVs can be charged simultaneously at a site. Thus, the **risk of congestion is reduced**, which, over and above **reducing the risk of charging time and queue formation, also reduces the risk of lost work,** which is heavily penalized. However, in our current case study, we do not observe a reduction of charging time as we have 2 CEVs work per site, and thus a single FCS having two ports is able to

charge both the CEVs simultaneously - i.e, the setup of our case study obviates the reduction of charging time which we would observe in general.

D. **Prove the benefits of MCSs for grid services, including but not limited to increased utilization of existing EV charging stations, higher renewable self-consumption, and emergency building services by integration with the world-renowned UCSD microgrid**
The UCSD campus operates a large and diverse EV charging network that appears to have spare capacity at many existing stations, suggesting an opportunity to increase utilization of these sunk infrastructure investments by using them to recharge MCSs when they are not serving light-duty EVs. In the proposed framework, the MCS routing and charging optimization will explicitly account for UCSD microgrid characteristics, including on-site PV generation and campus load profiles, so that MCSs can be scheduled to charge preferentially during periods of higher solar output and lower campus demand. In this way, MCSs are intended to function as controllable, mobile loads that can increase renewable self-consumption by time-shifting energy from daytime PV production to later use at construction sites. In addition, the project will build on initial demonstrations of MCSs as mobile backup power sources to design and evaluate operating strategies in which, when appropriately islanded, MCSs can support critical loads during grid outages. Together, these elements define a framework in which integration of MCSs with the world-renowned UCSD microgrid can provide grid services that include higher utilization of existing EV charging infrastructure, improved use of on-site renewable generation, and enhanced capability for emergency building support.

Objectives:

A. **Validate operations of MCSs for CEVs charging under regular construction activities for at least six months at multiple construction sites:** The first demonstration phase, involving regular construction activities using CEVs and MCSs, was conducted over one week (20–24 October 2025) at the RIMAC Arena of UC San Diego. The second demonstration phase, also at the RIMAC Arena, has been scheduled for a two-week duration and commenced on 2 February 2025; and is currently ongoing. We are also working with the campus to identify additional construction sites for demonstration and deployments.

B. **Reduce costs:** Demonstrate at least 10% peak demand reduction and charging time reduction (including travel, waiting & actual charging time) by using mobile charging stations versus the fixed charging station benchmark for construction equipment charging.

1. For the MCS with optimal scheduling we observe a **49% and 65% peak demand reduction** in Scenarios 1 and 2; respectively, as compared to the FCS.
2. If there are **more than two CEVs at a site which run out of charge together**, we will observe a congestion in charging in the case of FCS (which has two ports for charging CEVs), leading to **queue formation and increased waiting times, which in turn results in increased total time required for charging.** This congestion can also happen for the MCS (which also has two ports for charging CEVs), but if **multiple MCSs** are deployed, which can be **dynamically routed between the construction sites in response to demand**, more than two CEVs can be charged simultaneously at a site. Thus, the **risk of congestion is reduced**, which over and above **reducing the risk of charging time and queue formation, also reduces the risk of lost work** which is heavily penalized. However, in our case studies, we do not observe a reduction of charging time as we assume that only 2 CEVs work per site, and thus a single FCS having two ports is able to charge both the CEVs simultaneously - i.e, the setup of our case study obviates the reduction of charging time which we would observe in general.

C. **Reduce emissions: Demonstrate at least 50% carbon emission reduction by using electric construction equipment versus internal combustion construction equipment.**

The CO_2 emissions of the CEVs are 84-87 % lower than those of the diesel-powered ICE construction equipment. The lowering is due to lower marginal emissions of carbon in the (renewable heavy) CA grid as compared to carbon emissions due to diesel.

D. MCS-to-Grid Benefits: at least 5% increase in charging station utilization and renewable self-consumption, and validation of MCSs for building emergency services.

In this project, we will develop and apply an optimization and data-analytics framework that quantifies how routing and charging MCSs at UCSD's existing EV charging stations affect station utilization and renewable self-consumption on the UCSD microgrid. Using historical and demonstration-period data, we will establish baseline utilization and PV self-consumption and then evaluate MCS operation strategies that are explicitly co-optimized with campus load, PV generation, and charger availability, with the objective of achieving at least a 5% increase in both metrics. The same framework will be extended to analyze operating modes in which MCSs, when safely islanded, supply critical loads for selected buildings during emulated outage events, providing quantitative evidence of their capability as mobile emergency resources. Collectively, these studies will demonstrate how integrating MCSs with the UCSD microgrid can deliver measurable grid benefits, higher use of existing charging infrastructure, greater on-site renewable utilization, and validated strategies for building emergency support.

E. Technology Transfer: Test the MCS-to-CEV solution with different use cases, including but not limited to different MCSs battery types, CEV types, and construction locations.

The case studies investigate the performance under different operation strategies with different locations, working schedules, CEV number/types, and number of MCSs. As the MCS are typically oversized as compared to CEVs, the battery capacity of the MCSs are unlikely to have an effect on the performance in practical real-world deployment.

APPENDIX: DETAILED FORMULATION OF THE CONSTRAINTS IN THE OPTIMIZATION FRAMEWORK

A. Notation

The following is the notation employed throughout the optimization described hereon, with others being introduced as and when necessary. CEVs are indexed by $e \in E$ with E being the set of CEVs, nodes are indexed by $i, j \in N$ with N being the set of nodes, N^0 is the set of grid connection nodes (where MCS is connected to the grid), N' is the set of construction facility nodes (where CEVs work and are recharged from the MCS), S is the set of paths between nodes where MCS moves, implying $(i, j) \in S$, MCS are indexed by $m \in M$ with M being the set of MCS, and time is indexed by $t, t' \in T$ with T being the set of time. Note that $N = N^0 \cup N'$.

B. Constraints

a. Charging and discharging power constraints of the MCS and CEV

The constraints for charging and discharging power of the MCS and CEV are presented in (1) – (13). Equation (1) computes the total charging power of MCS m at time t, denoted as $P^{\mathrm{ch,tot}}_{m,t}$. $P^{\mathrm{ch,tot}}_{m,t}$ is represented by the sum of the charging power of MCS m at time t and node i, $P^{\mathrm{ch,MCS}}_{m,i,t}$ over all grid connection nodes N^0 as

$$P^{\mathrm{ch,tot}}_{m,t} = \sum_{i \in N^0} P^{\mathrm{ch,MCS}}_{m,i,t},\ \forall m \in M,\ \forall t \in T \tag{1}$$

Similarly, the total discharging power of MCS m at time t, $P^{\mathrm{dch,tot}}_{m,t}$, is determined in (2) as the sum of discharging power, $P^{\mathrm{dch,MCS}}_{m,i,t}$ of MCS m at time t and node i, over all construction nodes N' as

$$P^{\mathrm{dch,tot}}_{m,t} = \sum_{i \in N'} P^{\mathrm{dch,MCS}}_{m,i,t},\ \forall m \in M,\ \forall t \in T \tag{2}$$

The discharging power of MCS m at time t and construction node i, $P^{\mathrm{dch,MCS}}_{m,i,t}$, is equal to the sum of the power injected to CEV e at time t and node i by MCS m, $P^{\mathrm{MCS \to CEV}}_{m,i,e,t}$ over the set of CEVs E as indicated in (3). Note that one MCS can recharge multiple CEVs at the same time through using multiple plugs.

$$P^{\mathrm{dch,MCS}}_{m,i,t} = \sum_{e \in E} P^{\mathrm{MCS \to CEV}}_{m,i,e,t},\ \forall m \in M,\ \forall i \in N',\ \forall t \in T \tag{3}$$

Constraint (4) prevents charging of MCSs at the construction nodes N', while constraint (6) prevents discharging of MCSs at the grid connection nodes N^0.

$$P^{\mathrm{ch,MCS}}_{m,i,t} = 0,\ \forall m \in M,\ \forall i \in N',\ \forall t \in T \tag{4}$$

$$P^{\mathrm{dch,MCS}}_{m,i,t} = 0,\ \forall m \in M,\ \forall i \in N^0,\ \forall t \in T \tag{5}$$

The charging power of MCSs, $P^{\mathrm{ch,MCS}}_{m,i,t}$ is restricted by the maximum charging power rate, $\mathrm{CH}^{\mathrm{MCS}}$, as shown in (6). The binary variable $z_{m,i,t}$ checks the presence of MCS m at time t to node i, which in (6) happens to be the grid connection node.

$$P_{m,i,t}^{\mathrm{ch,MCS}} \leq \mathrm{CH}^{\mathrm{MCS}} z_{m,i,t}, \quad z_{m,i,t} \in \{0,1\},\ \forall m \in M,\ \forall i \in N^0,\ \forall t \in T \tag{6}$$

Similarly, (7) limits the discharging power of MCS m at time t and construction node i to the maximum discharging power rate of the plugs mounted on the MCS, $\mathrm{DCH}^{\mathrm{MCS}}$ as

$$P_{m,i,t}^{\mathrm{dch,MCS}} \leq \mathrm{DCH}^{\mathrm{MCS}} z_{m,i,t}, \quad z_{m,i,t} \in \{0,1\},\ \forall m \in M,\ \forall i \in N',\ \forall t \in T \tag{7}$$

where $z_{m,i,t}$ in (7) checks the presence of MCS m at time t in construction node i.
Constraint (8) ensures that the the power injected to CEV e at time t and node i by MCS m, $P_{m,i,e,t}^{\mathrm{MCS}\to\mathrm{CEV}}$ is limited by the discharging power of the MCS plug, $\mathrm{DCH}^{\mathrm{MCS,plug}} = \dfrac{\mathrm{DCH}^{\mathrm{MCS}}}{C_m^{\mathrm{plug}}}$, where C_m^{plug} is the number of discharging plugs on the MCS. Moreover, the binary variable $\varrho_{m,i,e,t}$ takes a value of 1 when CEV e connects to MCS m in time t at construction node i to initiate a charging session; otherwise, it takes a value of 0.

$$P_{m,i,e,t}^{\mathrm{MCS}\to\mathrm{CEV}} \leq \mathrm{DCH}^{\mathrm{MCS,plug}} \varrho_{m,i,e,t}, \quad \varrho_{m,i,e,t} \in \{0,1\},\ \forall m \in M,\ \forall i \in N',\ \forall e \in E,\ \forall t \in T \tag{8}$$

The charging power of CEVs are also limited by the charging power rate of each CEV, $\mathrm{CH}_e^{\mathrm{CEV}}$, as specified in (9). The binary variable $\mu_{i,e,t}$ takes a value of 1 when CEV e at time t and construction node i is charging; otherwise, it takes a value of 0. Note that (9) shows that one CEV can be charged from multiple MCS at a time using different plugs.

$$\sum_{m \in M} P_{m,i,e,t}^{\mathrm{MCS}\to\mathrm{CEV}} \leq \mathrm{CH}_e^{\mathrm{CEV}} \mu_{i,e,t}, \quad \mu_{i,e,t} \in \{0,1\},\ \forall i \in N',\ \forall e \in E,\ \forall t \in T \tag{9}$$

The power consumption of CEV e to perform work at construction node i at time t, $P_{i,e,t}^{\mathrm{work}}$ is limited by its working capacity M, as given in (10) ensuring that the CEV does not work and charge from the MCS at the same time.

$$P_{i,e,t}^{\mathrm{work}} \leq M\big(1 - \mu_{i,e,t}\big), \quad \mu_{i,e,t} \in \{0,1\},\ \forall i \in N',\ \forall e \in E,\ \forall t \in T \tag{10}$$

Constraint (11b) ensures that the CEV does not work during non-working hours of the day for the construction crew, represented by the time set T'.

$$P_{i,e,t}^{\mathrm{work}} = 0,\ \forall i \in N',\ \forall e \in E,\ \forall t \in T' \tag{11}$$

Constraint (12) ensures that $P_{i,e,t}^{\mathrm{work}}$ is limited by the power required to accomplish work at construction node i in time t, $R_{i,e,t}^{\mathrm{work}}$ as follows,

$$P_{i,e,t}^{\mathrm{work}} \leq R_{i,e,t}^{\mathrm{work}},\ \forall i \in N',\ \forall e \in E,\ \forall t \in T \tag{12}$$

Equation (13) calculates the missed work for each CEV e at construction node i and time t, based on the difference between the scheduled work and the performed work.

$$P_{i,e,t}^{\text{missed,work}} = R_{i,e,t}^{\text{work}} - P_{i,e,t}^{\text{work}},\ \forall i \in N',\ \forall e \in E,\ \forall t \in T \tag{13}$$

b. Energy consumption and state-of-energy constraints of the MCS and CEV

The energy consumption constraints of MCSs and CEVs are presented in (14) – (21). The energy consumption of MCS m in traveling from node i to node j, i.e., along path (i,j) at time t, $L_{m,i,j,t}^{\text{way}}$ is presented in (14) as,

$$L_{m,i,j,t}^{\text{way}} = k^{\text{way}} D_{i,j} x_{m,i,j,t},\ x_{m,i,j,t} \in \{0,1\},\ \forall m \in M,\ \forall (i,j) \in S,\ \forall t \in T \tag{14}$$

where k^{way} and $D_{i,j}$ represent the energy consumed per mile during the travel of MCS (kWh/mile), and, distance of path (i,j) (miles); respectively. $x_{m,i,j,t}$ is a binary variable that takes a value of 1 if the MCS m is traveling along path (i,j) at time t; and is 0 otherwise.

The total energy consumption for travel of the MCS m at time t, represented by $L_{m,t}^{\text{way,tot}}$ is given in (15) as the sum of $L_{m,i,j,t}^{\text{way}}$ over all possible paths.

$$L_{m,t}^{\text{way,tot}} = \sum_{(i,j)\in S} L_{m,i,j,t}^{\text{way}},\ \forall m \in M,\ \forall t \in T \tag{15}$$

Equation (16) relates each MCS's state of energy (SOE) dynamics to the stored energy at the previous time-step, and charging, discharging, and energy consumption during travel at the current time-step, considering the charging and discharging efficiencies of the MCS m represented as η^{ch}, and η^{dch}; respectively.

$$\text{SOE}_{m,t}^{\text{MCS}} = \text{SOE}_{m,t-1}^{\text{MCS}} + \left(P_{m,t}^{\text{ch,tot}} \Delta T \eta^{\text{ch}}\right) - \left(\frac{P_{m,t}^{\text{dch,tot}} \Delta T}{\eta^{\text{dch}}}\right) - L_{m,t}^{\text{way,tot}},\ \forall m \in M,\ \text{if } t \neq t_0 \tag{16}$$

COnstraint (17) sets minimum and maximum state of energy limits of MCSs ($\text{SOE}_{\text{min}}^{\text{MCS}}$ and $\text{SOE}_{\text{max}}^{\text{MCS}}$) to prevent overcharge and deep discharge, which can harm the MCS battery.

$$\text{SOE}_{\text{min}}^{\text{MCS}} \leq \text{SOE}_{m,t}^{\text{MCS}} \leq \text{SOE}_{\text{max}}^{\text{MCS}},\ \forall m \in M,\ \forall t \in T \tag{17}$$

Equation (18) ensures that the SOE of MCSs is equal to the initial value, $\text{SOE}_{m}^{\text{MCS,ini}}$, at the beginning of the test day, and returns to the same value at the end of the test day, to ensure no net charging/discharging during the day.

$$\text{SOE}_{m,t}^{\text{MCS}} = \text{SOE}_{m}^{\text{MCS,ini}},\ \forall m \in M,\ \text{if } t = t_0\ \vee\ t = t^{\text{end}} \tag{18}$$

For CEVs, the SOE dynamics is modeled in (19) based on the stored energy from the previous step, the charging energy that comes from the MCSs, and the energy discharged to accomplish work over all the construction nodes.

$$\mathrm{SOE}_{e,t}^{\mathrm{CEV}} = \mathrm{SOE}_{e,t-1}^{\mathrm{CEV}} + \sum_{m\in M}\sum_{i\in N'}\left(P_{m,i,e,t}^{\mathrm{MCS}\to\mathrm{CEV}} - P_{i,e,t}^{\mathrm{work}}\right)\Delta T,\ \forall e\in E,\ \text{if } t\neq t^0 \tag{19}$$

Equation (20) sets minimum and maximum state of energy limits of each CEV, $\mathrm{SOE}_{\mathrm{min},e}^{\mathrm{CEV}}$ and $\mathrm{SOE}_{\mathrm{max},e}^{\mathrm{CEV}}$, respectively; while the initial SOE values are specified in (21) to ensure that SOE of each CEV equals to $\mathrm{SOE}_{e}^{\mathrm{CEV,ini}}$ at both the beginning and end of the test day.

$$\mathrm{SOE}_{\mathrm{min},e}^{\mathrm{CEV}} \leq \mathrm{SOE}_{e,t}^{\mathrm{CEV}} \leq \mathrm{SOE}_{\mathrm{max},e}^{\mathrm{CEV}},\ \forall e\in E,\ \forall t\in T \tag{20}$$

$$\mathrm{SOE}_{e,t}^{\mathrm{CEV}} = \mathrm{SOE}_{e}^{\mathrm{CEV,ini}},\ \forall e\in E,\ \text{if } t=t_0\ \vee\ t=t^{\mathrm{end}} \tag{21}$$

c. Spatial movement and connection status of the MCSs

The spatial movement and connection status of the MCSs are expressed in (22) - (34). Equation (22) indicates that the number of CEVs connected to MCS m at construction node i and time t is limited by the number of available plugs on the MCS.

$$\sum_{e\in E}\varrho_{m,i,e,t} \leq C_{m}^{\mathrm{plug}},\quad \varrho_{m,i,e,t}\in\{0,1\},\ \forall m\in M,\ \forall i\in N',\ \forall t\in T \tag{22}$$

Constraint (23) ensures that the binary variable $\varrho_{m,i,e,t}$ indicating the connection status of CEV e to MCS m can take a value of 1, only if the binary parameter $A_{i,e}$, which represents the location of CEV e at construction node i, is equal to 1. Therefore, the MCS m can only charge CEV e at the construction node i where CEV e is located.

$$\varrho_{m,i,e,t} \leq A_{i,e},\quad \varrho_{m,i,e,t}\in\{0,1\},\ \forall m\in M,\ \forall i\in N,\ \forall e\in E,\ \forall t\in T \tag{23}$$

The binary variables $x_{m,i,j,t}$, $z_{m,i,t}$, $\gamma_{m,i,t}^{\mathrm{arr}}$ and $\sigma_{m,i,t}^{\mathrm{dep}}$ define spatio-temporal connections, controlling MCS movements between nodes based on scheduled working times of CEVs. As discussed previously in (14), the binary variable $x_{m,i,j,t}$ determines whether MCS m goes from node i to node j at time t. Constraint (24) guarantees that the number of arrivals at and departures from each node are equal for each MCS.

$$\sum_{i\in N}\sum_{t\in T}x_{m,i,j,t} = \sum_{i\in N}\sum_{t\in T}x_{m,j,i,t},\quad x_{m,i,j,t}\in\{0,1\},\ \forall j\in N,\ \forall (i,j)\in S \tag{24}$$

The binary variable $\gamma_{m,i,t}^{\mathrm{arr}}$ takes a value of 1 when MCS m arrives at node i at time t. Similarly, the binary variable $\sigma_{m,i,t}^{\mathrm{dep}}$ takes a value of 1 if MCS m departs from node i at time t. Thus, the presence of MCS m at node i at time t, denoted by the binary variable $z_{m,i,t}$ is derived based on the corresponding arrival and departure times, as described in (25).

$$\gamma_{m,i,t}^{\mathrm{arr}} - \sigma_{m,i,t}^{\mathrm{dep}} = z_{m,i,t} - z_{m,i,t-1},\quad \gamma_{m,i,t}^{\mathrm{arr}}\in\{0,1\},\ \sigma_{m,i,t}^{\mathrm{dep}}\in\{0,1\},\ z_{m,i,t}\in\{0,1\},\ \forall m\in M,\ \forall i\in N,\ \text{if } t\neq t^0 \tag{25}$$

Constraints (26) – (28) ensure that the MCSs are initially connected to a grid connection node at the beginning of the test day and can begin moving after the initial time interval t^0.

$$\sum_{m \in M, i \in N^0} z_{m,i,t} = |M|, \quad z_{m,i,t} \in \{0,1\}, \text{ if } t = t^0 \tag{26}$$

$$\gamma^{\text{arr}}_{m,i,t} = 0, \quad \gamma^{\text{arr}}_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall i \in N', \text{ if } t = t^0 \tag{27}$$

$$\sigma^{\text{dep}}_{m,i,t} = 0, \quad \sigma^{\text{dep}}_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall i \in N', \text{ if } t = t^0 \tag{28}$$

Constraint (29) avoids MCSs from visiting multiple nodes in the same time interval as shown below.

$$\sum_{i \in N} z_{m,i,t} \leq 1, \quad z_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall t \in T \tag{29}$$

Constraint (30) specifies that MCS m must leave node i in time t to arrive at node j in time t', where $t^{\text{way}}_{i,j}$ is the time taken to travel from node i to j.

$$x_{m,i,j,t'} \leq \sigma^{\text{dep}}_{m,i,t}, \quad x_{m,i,j,t'} \in \{0,1\}, \ \forall m \in M, \ \forall (i,j) \in S, \ \forall t \in T, \ \forall t' \geq t + t^{\text{way}}_{i,j} \tag{30}$$

When the binary variable $\gamma^{\text{arr}}_{m,j,t}$ equals 1, indicating that MCS m reaches node j in time t, the condition outlined in (31) must hold to ensure conservation of flow (travel) of MCS m to node j from all other nodes i connected by path (i,j).

$$\gamma^{\text{arr}}_{m,j,t} = \sum_{i \in N} x_{m,i,j,t}, \quad \gamma^{\text{arr}}_{m,j,t} \in \{0,1\}, \ \forall m \in M, \ \forall (i,j) \in S, \ \forall t \in T \tag{31}$$

To avoid any overlap where an MCS both arrives and departs from the same node within the same time, constraint (32) is enforced.

$$\gamma^{\text{arr}}_{m,i,t} + \sigma^{\text{dep}}_{m,i,t} \leq 1, \quad \gamma^{\text{arr}}_{m,i,t} \in \{0,1\}, \ \sigma^{\text{dep}}_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall i \in N, \ \forall t \in T \tag{32}$$

Equations (33) and (34) limit MCS m to a single visit to each construction node per day, reducing unnecessary traffic and mitigating congestion issues, especially in constrained environments, such as university campuses or narrow urban streets.

$$\sum_{t \in T} \gamma^{\text{arr}}_{m,i,t} \leq 1, \quad \gamma^{\text{arr}}_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall i \in N' \tag{33}$$

$$\sum_{t \in T} \sigma^{\text{dep}}_{m,i,t} \leq 1, \quad \sigma^{\text{dep}}_{m,i,t} \in \{0,1\}, \ \forall m \in M, \ \forall i \in N' \tag{34}$$